higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

AUTOMOTIVE REPAIR AND MAINTENANCE
NQF LEVEL 3

21 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1
1.1.1 D
1.1.2 J
1.1.3 H
1.1.4 A
1.1.5 E
1.1.6 K
1.1.7 C
1.1.8 F
1.1.9 I
1.1.10 B

(10 × 1) (10)

1.2
- Heating of the combustion chambers is necessary because it is a compression ignition engine.
- Compression and heat are used to ignite the fuel.
- When the engine is cold, pre-heating is used to ensure proper combustion of the fuel. (3)

1.3
- Back-leakage test
- Seat-leakage test
- Opening-pressure test/Break pressure
- Spray-pattern test (Any 3 × 1) (3)

1.4 It is the procedure for removing air from the fluid in the brake system. (1)

1.5 1.5.1
- Out of round brake drums
- Warped brake discs
- Hard spots on the braking surface of the drums or discs
- Loose callipers
- Loose backing plates (Any appropriate answers 2 × 1)

1.5.2
- Incorrect adjustment of brake shoes
- Incorrect adjustment of parking brake/hand brake
- Incorrect adjustment of brake master cylinder push-rod
- Loose brake lining (Any appropriate answers 2 × 1)

(2 × 2) (4)

1.6
- Remove the brake drum
- Remove the brake shoes in the correct order
- Remove the brake pipe at the wheel cylinder
- Remove the wheel cylinder (4)

1.7
- It locks the output gear with the output shaft
- It prevents the gears from grinding during engagement
- It synchronises the speed of the gears for smooth selection (3)

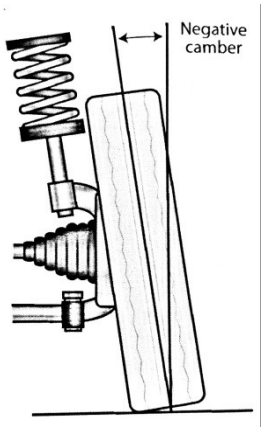
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1.8 A – Excessive toe contact
B – Excessive heel contact
C – The desired contact (3)

[31]

QUESTION 2

2.1



(3)

2.2
- To reduce tilting of the wheel when moving up and down
- To reduce changes in the wheel track
- To reduce tyre scrub as it moves up and down
- To allow the wheels to remain upright when cornering
- It also fits in the space allowed with the upper arm next to the engine bay

(Any 3 × 1) (3)

2.3 2.3.1
- The shock absorbers prevent continues bouncing of the wheel after up or down movement
- They smooth out up and down movement
- They provide a smooth ride
- They prevent excessive wear of other components like tyres, bushes

2.3.2
- The coil springs support the weight of the vehicle
- They allows up or down movement of the suspension

(2 × 2) (4)

2.4
- Too much inward or outward tilt of the wheel
- The wheel position is too much to the front or back
- Abnormal tyre wear on the inside or outside of the tyre
- Steering pulling to one side on a level road

(Any ONE appropriate answer) (1)

2.5
- Knock-on weights
- Clip-on weights
- Stick-on weights (Any 2 × 1) (2)

2.6
- Check for physical damage/dents
- Check for mud or dirt that will affect the balancing
- Check whether all old weights have been removed

(Any appropriate answer) (3)

2.7 Dynamic balancing (1)

2.8
- Bouncing of the wheels
- Vibration on the steering wheel
- Uneven tyre wear
- Premature wear on suspension parts
- Premature wear on steering parts
- Uncomfortable ride (Any appropriate answer 4 × 1) (4)

[21]

QUESTION 3

3.1 3.1.1
- The solenoid
- The field-coils/stator
- The armature (Any 2 × 1) (2)

3.1.2 A – Brush spring
B – Fork
C – Commutator (3)

3.2 3.2.1
- Pinion not engaging
- Pinion teeth stripped
- Pinion clutch faulty
- Flywheel ring-gear stripped
- Bendix dry or sticky (Any ONE appropriate answer) (1)

3.2.2
- Solenoid faulty
- Supply wire from solenoid to starter motor broken
- Brushes worn
- Commutator faulty
- Armature faulty (Any ONE appropriate answer) (1)

3.3 3.3.1 The battery light will light up (1)

3.3.2
- The battery voltage will run down
- Low battery voltage can affect the ignition system
- Low battery voltage can affect the electronic system
- The vehicle could cut out (Any ONE appropriate answer) (1)

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3.4
- Loosen the alternator mounting bolts
- Loosen the adjusting bracket bolts
- Pull the alternator away from the engine with a lever to tension the v-belt
- Tighten the adjusting bracket and alternator mounting bolts (4)

3.5 Stator windings (1)

3.6 3.6.1 Vacuum advance
3.6.2 Mechanical advance/Centrifugal advance
(2 × 1) (2)

3.7 3.7.1 Timing light
3.7.2 Feeler gauge
3.7.3 Multimeter
3.7.4 Vacuum pump
(4 × 1) (4)

3.8
- Air-flow meter
- Manifold pressure sensor
- MAP sensor
- Air temperature sensor
- Coolant temperature sensor
- Knock sensor
- Throttle position sensor
- Intake air temperature sensor (Any appropriate answer 3 × 1) (3)

3.9
- Coil
- Distributor cap
- High tension wires
- Rotor
- Spark plugs (Any ONE) (1)

[24]

QUESTION 4

4.1 4.1.1 A – Camshafts
C – Cam followers
B – Valves (4)

4.1.2
- Piston diameter
- Ring groove depth
- Ring groove clearance
- Gudgeon-pin bore
- Piston out of round
- Piston skirt collapse (Any 2 × 1) (2)

4.1.3
- The crankshaft
- The camshaft (2)

4.1.4
- Send the part for engineering or machining work
- Receive the correct part back from the engineers
- Inspect the work done on the part (3)

4.2 4.2.1 Ring squeezer
4.2.2 Valve spring compressor/valve lifter
4.2.3 Torque wrench
(3 × 1) (3)

4.3
- Place the engine on a proper stand
- Close ALL the exposed holes to prevent foreign matter from entering
- Lubricate parts to prevent them from rusting in storage
- Wrap components in protective material to prevent damage
- Do not store near flammable liquid
- Store in a cool, dry place (Any appropriate answers 5 × 1) (5)

4.4
- Check for oil leaks
- Check for water leaks
- Check the idling
- Check the ignition timing
- Check electrical connections
- Check mechanical connections
- Check mounting bolts and nuts
- Check engine performance (Any appropriate answers 5 × 1) (5)

[24]

TOTAL: 100

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